

Bridging Gaps in Heliospheric Radio Data Analyses

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Scientific rationale

Scientific value of heliospheric radio observations:

- Ground- and space-based observations → combined to probe entire space between Sun + Earth
- Diagnostic tool for accelerated particles + heliospheric environment (e.g. turbulence)
- Multi-instrument + multi-vantage observations → semi-continuous monitoring
- Valuable addition to multi-wavelength studies
- Frequent events → vast amounts of data

BUT: broader scientific use remains limited
→ several barriers for (new) users

User challenges:

- Limited awareness of instruments, available data products, and access methods
- Quick-look plots not always available/easy to find
- Need to handle diverse range of complex data formats + metadata
- Insufficient documentation, or difficult to find

AIM: provide solutions to improve data usability

Open-source data container

- Initiated the development of an **open-source modular data container**
- **Collated the use cases** for spectroscopic heliospheric radio data → guidelines for the container development

Objectives of the standardised data container:

- Handle data of various formats from various radio instruments
→ **streamline access and analysis** of heliospheric radio data
- Be integrated into **community-driven open software**
→ SunPy ecosystem in Python

Areas covered by the user wish list:

- Data visualisation
- Basic operations
- Multi-instrument operations
- Operations for scientific analysis
- Cross-dataset and larger-scale studies

* Have a look at our website
for more details!

Instrument catalogue

- Curated a catalogue to mitigate the difficulties in identifying available heliospheric radio instruments and accessing their data.

→ **An open-source community resource**
(anyone can contribute!)

The published catalogue currently provides:

1. Key instrument and data information (name, frequency range, types of data products, output data formats, geographic location, window of operation)
2. Links to official data sources and further information
3. Links to quick-look plots
4. Public software tools for reading the data
5. Other useful information and instrument-specific contacts

It includes:

- Ground- and space-based instruments
- Active and decommissioned instruments



instrument catalogue

Software + Notebooks

Surveying + comparing existing analysis tools:

- Tested capabilities:
 - download, read, and plot spectroscopic radio data
 - handle and combine data from various sources and of various data formats
- Identified + documented aspects to improve
- Implemented + published several improvements
→ Example contributions: SciQLop (LOFAR support); SunPy/radiospectra

Open-source, example (Jupyter) notebooks:

- Scope: download, read, and visualise diverse radio datasets in Python
- Provide guidance on how to bypass identified challenges

Available Jupyter notebooks (on GitHub):

- software specific (e.g. SciQLop, PySPEDAS, MASER, radiospectra)
- data-format specific (e.g. CDF, FITS)
- provider specific (e.g. CDAWeb)
- instrument specific (e.g. SoLO, PSP, STEREO, WIND, LOFAR, OVRO-LWA, ORFEES, RSTN, etc.)

* Have a look at our
GitHub repo!

Requirements + Community roadmap

- Produced a **community-driven roadmap** for interoperable heliospheric radio data
- **Identified and documented the requirements** for a standardised approach to heliospheric radio data → considered both ground- and space-based instruments
- **Provided recommendations** for:
 - (1) instrument teams + data providers,
 - (2) software and tool developers,
 - (3) users

Container, File format, and Software requirements – areas covered:

- Data model
- File format
- Construction
- Core operations
- Heterogeneous and multi-resolution data
- Search and retrieval
- Broader integration/Interoperability
- Documentation and quality

Metadata requirements – areas covered:

- Identification
- Time
- Frequency
- Signal and Calibration
- Data quality
- Position and pointing

* Have a look at our website
for more details!

Getting involved + Additional details

Reach out + get involved!

→ We encourage contributions, feedback, and suggestions.

Website



GitHub



Chat room



- Dedicated website: lc-radio-workshop.github.io
- GitHub repository: github.com/LC-radio-workshop
- Chat room: matrix.to/#/#lc-radio-workshop-community-space:matrix.org
- Lorentz Center page: lorentzcenter.nl/bridging-gaps-in-heliospheric-radio-data-analyses.html

* This work began as part of a Lorentz Center workshop held from 18-22 May 2026.
* **More details will be published in a white paper:** Chrysaphi et al. (in prep.)



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